# Lucas McLaren

lucasgmclaren@gmail.com • +1 (650) 670-6516

## Education

**Bachelor of Science in Mechanical Engineering – Magna Cum Laude** *Chico State University* - GPA: 3.76

Relevant Coursework: Fluid Dynamics – Energy Systems – Strength of Materials – Advanced Manufacturing

**Tau Beta Pi -** Engineering Honors Society Member

## Experience

**Society of Automotive Engineers** Chico, CA

**SAE Baja Club Officer** August 2024 – May 2026

- Leadership and management of gearbox sub team
- Engineering on a large scale working in groups
- Multiple subgroups collaborating to create one cohesive vehicle
- Automotive industry simulating experience
- Design, validation, manufacturing, composite experience, assembly, and testing

**Construction Management** San Francisco, CA

**Southland Industries Project Engineering Internship** June 2025 – August 2025

- MEP experience on the largest healthcare construction project in California
- Teamwork and team development
- Documentation gathering and organization
- Start to finish HVAC design and estimation project
- Communication between several departments and multiple companies

## Projects

**Academic**

**4 Wheel Drive RC Car (1st Place Machine Design)** January 2024 – May 2024

- Detailed design and evaluation in a comprehensive design report
- Advanced manufacturing processes used in a professional workshop
- Complex geometry 3D printed carbon fiber frame
- Integrated off-the-shelf components with custom drive train

**Automated Microscopic Flaw Inspection System (1st Place Capstone)** September 2025 – May 2026

- Project Manager of a four-person team
- Sponsored by Lawrence Livermore National Laboratory
- Start to finish product development for fusion research technologies
- Budget analysis and cost evaluation throughout project
- Working with sponsors for requirement definitions and design reviews

**Personal**

**Moped Rebuild** June 2022 – August 2023

- Modifying existing chassis for redesign
- Manufacturing custom body panels
- Wiring custom brake lights, turn signals, and headlights
- Sourcing parts and working with a group

**Mini Go-kart Upgrade** July 2024 – August 2024

- Fabricating a custom steel engine mount in a professional grade workshop
- Measuring and connecting custom chain and gears for increased torque
- New battery and motor holding tray and protective frame

## Skills & Interests

**Programs:** SolidWorks' CSWA Certified Associate – Java – Python – MATLAB – LabVIEW – Excel – Logger Pro

**Strengths:** Leadership – Group Collaboration – Mechanical Analysis – Mechanical Design – Manual Machining
Custom Fabrication – Manufacturing Capabilities – Automotive Experience – Quick Learning – Improvising in the Field